

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5 ASSESSMENT TOOL 1 — RESEARCH PAPER REVIEW / LITERATURE SURVEY

"Intelligent Public Health Surveillance and Disease Outbreak Prediction System"

Course Code: CSA1013 **Course Title:** Software Engineering

CO Assessed: CO5

Assessment: Assessment Tool 1 – Research Paper Review / Literature Survey

Weightage: 20% **Total Marks:** 25

Student Name: **SHEIK MOHAMED M**

Reg: **192511145**

Date: _____

Duration: 60 minutes

CO5 Statement: *Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly.*

Code of Conduct

I SHEIK MOHAMED M (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Assigned Problem Statement

19. Intelligent Public Health Surveillance and Disease Outbreak Prediction System

Industry Context

Public health agencies require continuous disease surveillance to detect outbreaks and implement preventive measures. Artificial Intelligence, Big Data Analytics, Geographic Information Systems (GIS), the Internet of Things (IoT), and cloud-based healthcare platforms enable early detection of infectious diseases and support evidence-based public health decision-making.

Problem Statement

Disease outbreaks spread rapidly due to delayed detection, fragmented healthcare reporting systems, and inadequate epidemiological analysis. Existing surveillance systems often lack real-time monitoring and predictive capabilities. An intelligent disease surveillance platform should collect healthcare data from hospitals, laboratories, and public health agencies, analyze disease trends using AI algorithms, identify potential outbreak hotspots, and provide early warning notifications. The system should also support visualization, resource planning, and policy decision-making to minimize disease transmission and improve community health.

Objectives

1. Collect real-time public health data.
2. Predict disease outbreaks using AI.
3. Identify outbreak hotspots through GIS.
4. Generate early warning alerts.
5. Monitor disease transmission trends.
6. Support healthcare resource planning.
7. Provide epidemiological analytics dashboards.
8. Improve public health decision-making.

Expected Outcomes

1. Early disease outbreak detection.
2. Improved public health surveillance.
3. Reduced disease transmission.
4. Better healthcare resource allocation.
5. Faster emergency response.
6. Enhanced epidemiological analysis.
7. Improved healthcare planning.
8. Increased community health protection.

Executive Summary

This report presents a literature survey conducted in fulfillment of CO5 Assessment Tool 1 for the course Software Engineering (Course Code: CSA10), on the assigned problem statement "Intelligent Public Health Surveillance and Disease Outbreak Prediction System." The objective of the assignment is to evaluate emerging technologies and their societal impact by surveying existing research relevant to a real-world software-engineering problem — in this case, the design of an AI-, GIS-, IoT-, and cloud-enabled platform for early detection and prediction of infectious-disease outbreaks.

Ten peer-reviewed and reputable open-access research contributions, published between 2020 and 2026, were identified through a structured search of PubMed/PMC, ScienceDirect, Springer, MDPI, ResearchGate, and IJRASET. The selected papers were reviewed individually and then compared against a common set of parameters — methodology, dataset/tools used, and key findings — before being synthesized into a consolidated discussion of limitations, an identified research gap, and a proposed set of future research directions aligned with each objective of the assigned problem statement.

The survey finds that individual technology dimensions — AI/ML-based prediction, GIS-based spatial hotspot detection, IoT/cloud-based real-time acquisition, and digital/social-media-based early warning — have each been independently validated with measurable benefit (for example, up to 24 days of earlier outbreak detection using EHR-based machine-learning models, and multi-week lead time using digital-surveillance signals). However, no reviewed study integrates all four dimensions into a single, empirically validated, real-time, decision-support platform. This is identified as the central research gap, and Section 8 of this report proposes a six-part set of future research directions — spanning data integration, explainable AI, fine-grained GIS analytics, edge-enabled IoT pipelines, standardized evaluation benchmarks, and administrator-facing dashboards — intended to close this gap and directly satisfy the eight objectives and eight expected outcomes listed in the assigned problem statement.

1. Introduction

Infectious disease outbreaks remain among the most disruptive threats to public health, economic stability, and social order, as demonstrated repeatedly by seasonal influenza, dengue, cholera, and, most recently, the COVID-19 pandemic. Conventional public health surveillance depends heavily on manual case reporting by hospitals and laboratories, a process that is inherently slow, fragmented across administrative boundaries, and reactive rather than predictive. By the time a cluster of cases is officially confirmed and reported through traditional channels, an outbreak may already have spread well beyond its original point of origin, reducing the effectiveness of containment measures.

The convergence of Artificial Intelligence (AI), Big Data Analytics, Geographic Information Systems (GIS), the Internet of Things (IoT), and cloud computing has opened new possibilities for building intelligent, proactive disease surveillance ecosystems. Machine learning and deep learning models can identify subtle patterns in heterogeneous datasets — clinical records, laboratory results, mobility data, climate variables, wastewater signals, and even social media chatter — that are imperceptible to manual analysis. GIS-based spatial statistics can pinpoint geographic hotspots and transmission corridors, while IoT-enabled sensors and wearable devices allow

continuous, low-latency monitoring of physiological indicators at population scale. Cloud infrastructure, in turn, provides the elastic storage and computing capacity needed to process and correlate these large, fast-moving data streams in near real time.

This report presents a structured literature survey undertaken as part of the CO5 Assessment Tool 1 exercise for the assigned problem statement, "Intelligent Public Health Surveillance and Disease Outbreak Prediction System." The survey examines ten peer-reviewed and archival research contributions spanning four complementary technology dimensions relevant to the assigned problem: (i) AI/machine-learning-driven outbreak prediction, (ii) GIS-based spatial epidemiology and hotspot detection, (iii) IoT- and cloud-based real-time health data acquisition, and (iv) digital/social-media-based early warning systems. For each dimension, representative studies are reviewed in terms of their objectives, methodologies, datasets and tools, and key findings. The reviewed works are then compared using a common set of parameters, followed by a consolidated discussion of limitations, an identification of the research gap relevant to the assigned problem, and a set of future research directions and proposed system improvements.

The remainder of this report is organized as follows. Section 2 describes the literature search and paper-selection methodology. Section 3 provides conceptual background on the four enabling technology pillars. Section 4 presents the individual review of each selected research paper. Section 5 provides a comparative analysis of the reviewed literature in tabular form. Section 6 discusses common limitations and challenges identified across the literature. Section 7 identifies the research gap with respect to the assigned problem statement. Section 8 proposes future research directions and an outline of an improved system architecture. Section 9 concludes the report, and Section 10 lists the references used, formatted in IEEE citation style.

2. Literature Search and Selection Methodology

In line with the assessment instructions to identify and select relevant research papers from reputed journals, conferences, and scholarly databases, a structured search process was followed for this survey.

2.1 Databases Searched

- PubMed / PubMed Central (PMC)
- ScienceDirect
- Springer Link
- MDPI open-access journals
- ResearchGate and Academia.edu (for open-access preprints and archival copies)
- IJRASET and other UGC-recognized engineering research repositories

2.2 Search Terms Used

Combinations of the following keyword strings were used across the databases listed above: "disease outbreak prediction AND machine learning"; "public health surveillance AND artificial intelligence"; "GIS hotspot

detection AND epidemiology"; "IoT healthcare surveillance AND cloud computing"; "digital epidemiology AND social media early warning"; and "spatiotemporal disease cluster detection".

2.3 Inclusion and Exclusion Criteria

Papers were included if they (a) directly addressed disease outbreak prediction, public-health surveillance, or a closely related enabling technology named in the assigned problem statement (AI, GIS, IoT, cloud, or digital/social surveillance); (b) were published in a peer-reviewed journal, an indexed conference, or a reputable open-access repository; and (c) provided sufficient methodological detail to allow extraction of objectives, methodology, dataset/tools, and findings. Papers were excluded if they were purely opinion pieces without a described methodology, were inaccessible in full text, or fell entirely outside the scope of infectious-disease surveillance (e.g., papers focused solely on chronic, non-communicable disease management).

2.4 Selection Outcome

Following this process, ten research papers were selected for detailed review, spanning the period from 2020 to 2026 and covering all four technology dimensions named in the assigned problem statement. This spread was intentionally chosen so that the comparative analysis in Section 5 could evaluate the assigned problem from multiple complementary angles rather than a single narrow technique.

3. Enabling Technologies: Conceptual Background

Before reviewing the individual research papers, it is useful to briefly outline the four core technology pillars named in the assigned problem statement, since the selected literature is organized around them and the comparative analysis in Section 5 refers back to these categories.

3.1 Artificial Intelligence and Machine Learning for Outbreak Prediction

Machine learning models used for outbreak prediction generally fall into two families. Supervised models — such as logistic regression, random forests, gradient-boosted trees, and recurrent architectures like Long Short-Term Memory (LSTM) networks — are trained on historically labeled data (confirmed diagnoses, confirmed outbreaks) to forecast future case counts or outbreak probability. Unsupervised and anomaly-detection models — such as clustering algorithms and isolation forests — instead look for statistical deviations from expected baselines without requiring labeled outbreak events, making them useful for detecting novel or previously unseen pathogens. Natural Language Processing (NLP) further extends AI-based surveillance by extracting epidemiological signals from unstructured text such as clinical notes, news articles, and social-media posts.

3.2 Geographic Information Systems (GIS) and Spatial Epidemiology

GIS-based techniques allow disease case data to be analyzed not just as counts over time but as patterns over space. Spatial autocorrelation statistics such as Global and Local Moran's I quantify whether nearby locations have similar case rates, while the Getis-Ord G_i^* statistic identifies statistically significant clusters of high values (hotspots) or low values (coldspots). Kernel Density Estimation (KDE) produces smooth risk-density surfaces from discrete case points, and the Knox test evaluates whether cases are clustered in both space and time

simultaneously — a strong indicator of person-to-person or vector-borne transmission. When combined with remote-sensing covariates (e.g., land cover, vegetation indices, water bodies), these methods can also generate environmental risk maps that anticipate where future cases are likely to emerge.

3.3 Internet of Things (IoT) and Cloud-Based Health Data Acquisition

IoT extends surveillance beyond periodic clinical reporting by enabling continuous, sensor-driven data capture — from wearable devices (heart rate, temperature, oxygen saturation), smart thermometers, and environmental sensors, to wastewater monitoring stations. The Internet of Medical Things (IoMT) refers specifically to the subset of IoT devices used in healthcare contexts. Because these devices generate large, continuous data streams, cloud computing platforms are typically used to store, process, and analyze the resulting big data, often using distributed processing frameworks such as Hadoop MapReduce or streaming platforms such as Apache Kafka and Spark Streaming. Edge and fog computing architectures are increasingly used alongside the cloud to perform initial data filtering and inference closer to the data source, reducing latency and bandwidth demand.

3.4 Digital and Social-Media-Based Early Warning

Digital epidemiology uses non-clinical digital traces — search-engine query volumes (e.g., Google Trends), social-media posts, and participatory self-reporting platforms — as proxy indicators of disease activity in a population. Because these signals are generated by the public in near real time, and do not depend on a patient visiting a clinician or a laboratory confirming a diagnosis, they can sometimes reveal an emerging outbreak days to weeks before it appears in official statistics. However, this approach requires careful statistical validation against official case data to distinguish a genuine health signal from unrelated media-driven spikes in discussion volume.

4. Individual Review of Selected Research Papers

4.1 [P1] Ekundayo, F. (2024) — Using Machine Learning to Predict Disease Outbreaks and Enhance Public Health Surveillance

Source: World Journal of Advanced Research and Reviews, 24(3), 794–811.

Objective: To examine how machine learning techniques can be integrated with heterogeneous public-health data sources to improve the accuracy and timeliness of disease outbreak prediction.

Methodology: A conceptual and literature-based review of supervised learning (regression, ensemble classifiers) and unsupervised clustering approaches applied to electronic health records (EHRs), social media streams, climate variables, and genomic sequence data. The paper also discusses explainable AI (XAI) techniques such as attention mechanisms and Shapley values for improving model transparency.

Dataset / Tools: Synthesized from multiple secondary sources: EHR repositories, genomic databases, and social-media feeds; no single primary dataset.

Key Findings: Supervised models were found effective for forecasting influenza-like illness trends, while unsupervised clustering was useful for anomaly detection indicative of emerging pathogens. The study stresses that combining diverse data streams substantially improves prediction accuracy compared with single-source models, and highlights data governance, model interpretability, and cross-sector collaboration as prerequisites for real-world deployment.

Limitation: Being a review rather than an empirical study, it does not report quantitative accuracy figures for a specific deployed system; ethical and privacy governance challenges are only discussed qualitatively.

Relevance to Assigned Problem: *Directly supports Objective 2 (predict disease outbreaks using AI) by showing that fusing EHR, climate, and social data with supervised/unsupervised ML improves prediction accuracy, and its emphasis on explainable AI informs the trust requirements of Objective 8 (improve public health decision-making).*

4.2 [P2] SciRP Journal Study (2026) — Artificial Intelligence-Driven Approaches to Disease Surveillance and Outbreak Prediction

Source: Scientific Research Publishing (SCIRP), Open-Access Health Informatics Journal.

Objective: To review how AI techniques — machine learning, natural language processing (NLP), and deep learning — can overcome the reporting delays and under-detection inherent in conventional surveillance systems.

Methodology: Structured literature review classifying AI applications into four functional categories: automated case detection, syndromic surveillance, real-time anomaly detection, and predictive outbreak modeling.

Dataset / Tools: Aggregated evidence drawn from published case studies of automated surveillance systems; not tied to one dataset.

Key Findings: AI-enabled automation of case detection and anomaly flagging reduces the lag between disease onset and official reporting. Predictive modeling using historical and real-time data supports earlier intervention. The review emphasizes that AI adoption must be paired with robust validation to avoid alert fatigue from false positives.

Limitation: Provides limited discussion of computational cost, scalability testing, or interoperability with legacy hospital information systems.

Relevance to Assigned Problem: *Supports Objective 4 (generate early warning alerts) by demonstrating how AI-based automated case detection and anomaly flagging shorten the delay between disease onset and confirmed reporting, a core requirement of the assigned surveillance platform.*

4.3 [P3] PMC Study Group (2024–2025) — AI-Enabled Diagnostic Prediction within Electronic Health Records to Enhance Biosurveillance and Early Outbreak Detection

Source: PubMed Central (PMC).

Objective: To develop and validate a machine-learning model that predicts, within a 24-hour window, the likelihood of a patient being diagnosed with an infectious gastrointestinal, respiratory, or neurological condition, and to combine these predictions with spatiotemporal outbreak-detection algorithms.

Methodology: Retrospective cohort analysis using supervised ML models trained on historical EHR diagnostic codes; high-confidence predictions were fused with confirmed diagnoses and analyzed with spatiotemporal cluster-detection statistics.

Dataset / Tools: 4.5 million de-identified patient records spanning 2010–2022, drawn from a large regional healthcare system.

Key Findings: Among diseases with five or more recorded outbreaks between 2014 and 2022, the ML-augmented approach detected 33.3% of outbreaks earlier than conventional surveillance (41 of 123 outbreaks), with lead times of 1 to 24 days and an average of only 1.33 false-positive outbreak alerts per year.

Limitation: Performance is highly dependent on the availability of sufficient disease-specific historical training data; diseases with sparse historical records showed weaker predictive lead time.

Relevance to Assigned Problem: *Provides the strongest empirical evidence for Objective 2, showing measurable earlier detection (up to 24 days) using real EHR data at scale — a benchmark that any proposed intelligent surveillance system for this assignment should aim to match or exceed.*

4.4 [P4] Ajeema Arsi, A. et al. (2025) — Harnessing Machine Learning and AI Techniques for Accurate Disease Outbreak Forecasting

Source: International Journal for Research in Applied Science and Engineering Technology (IJRASET), DOI: 10.22214/ijraset.2025.73613.

Objective: To investigate the integration of AI, machine learning, and deep learning approaches for early and precise disease outbreak forecasting using diverse data sources.

Methodology: Comparative survey of forecasting models applied to medical records, meteorological data, and social signals, with discussion of a proposed real-time outbreak-monitoring system design.

Dataset / Tools: Composite of medical records, weather/climate datasets, and social-media-derived signals reported across the surveyed studies.

Key Findings: The combination of AI/ML/DL techniques with multi-modal data (clinical + environmental + social) consistently outperformed single-source statistical forecasting baselines. The authors argue that continued technological development and cross-sector collaboration between healthcare and research institutions are essential to operationalize such systems.

Limitation: Data privacy, data quality inconsistency, and limited interpretability of deep-learning forecasts are identified as open and unresolved issues.

Relevance to Assigned Problem: *Reinforces Objective 2 by confirming that multi-modal AI/ML/DL fusion (clinical + environmental + social) outperforms single-source forecasting, and its discussion of data-privacy and interpretability challenges is directly relevant to the governance dimension of Objective 8.*

4.5 [P5] MDPI Study (2025) — AI-Assisted Real-Time Monitoring of Infectious Diseases in Urban Areas ("SmartHealth-Track")

Source: MDPI Mathematics, 13(12), 1911.

Objective: To design a scalable AI-based real-time surveillance architecture, named SmartHealth-Track, that fuses IoT-enabled monitoring, wearable health data, smart-pharmacy analytics, wastewater surveillance, and NLP-based text mining for early outbreak detection in dense urban populations.

Methodology: System design and simulation combining LSTM-based time-series forecasting, logistic regression for outbreak-probability estimation, isolation-forest-based anomaly detection, and NLP for extracting epidemiological signals from unstructured sources.

Dataset / Tools: Simulated/composite urban health data streams: wearable sensor feeds, pharmacy sales records, and wastewater surveillance indicators.

Key Findings: The multi-source fusion architecture improved both the accuracy and timeliness of outbreak prediction relative to single-source models, and the authors highlight edge-compatible, lightweight model deployment as a means of reducing latency in dense urban networks.

Limitation: Interoperability across heterogeneous device ecosystems, data-privacy compliance, and minimizing false-positive alerts remain unresolved implementation challenges noted by the authors themselves.

Relevance to Assigned Problem: *Closely aligned with Objectives 1, 4, and 5 (real-time data collection, early warning alerts, and monitoring transmission trends), as SmartHealth-Track's fusion of wearable, pharmacy, and wastewater data mirrors the multi-source real-time architecture the assigned problem statement calls for.*

4.6 [P6] ResearchGate Study (2021) — GIS-Based Spatial Analysis and Hotspot Detection of COVID-19 Outbreak

Source: ResearchGate (open-access preprint/publication).

Objective: To characterize the global spatial distribution of COVID-19 incidence and fatality and to identify statistically significant high-risk and low-risk geographic clusters.

Methodology: Calculation of cumulative incidence rate (CIR) and cumulative fatality rate (CFR) per country; spatial-pattern analysis using Global Moran's I; hotspot detection using the Getis-Ord G_i^* statistic.

Dataset / Tools: Global, country-level COVID-19 case and fatality data collected from the outbreak onset through January 2021.

Key Findings: Significant spatial clustering of COVID-19 fatality was identified, with Yemen, Mexico, and Ecuador showing the highest case-fatality clusters, while Andorra, Gibraltar, and Montenegro showed the highest incidence-rate clusters. The study demonstrates that classical spatial-statistics techniques remain effective for macro-level outbreak monitoring even without individual-level patient data.

Limitation: Country-level aggregation masks sub-national (city/village-level) variation and cannot support the fine-grained hotspot alerts needed for local public-health intervention.

Relevance to Assigned Problem: *Directly supports Objective 3 (identify outbreak hotspots through GIS) by demonstrating that classical spatial statistics can reliably flag country-level high-risk clusters, providing a macro-scale template for the hotspot-identification component of the assigned system.*

4.7 [P7] Kolkata/West Bengal GIS Study — Spatiotemporal Pattern and Hotspot Detection of Malaria and Dengue Using Spatial Analysis and GIS: An Approach to Medical GIS

Source: ResearchGate.

Objective: To map the spatiotemporal diffusion pattern of dengue and malaria outbreaks at fine spatial granularity and to build an environmental risk-prediction model.

Methodology: Daily case-mapping over the outbreak period; spatio-temporal clustering using the Knox test; an environmental risk model derived from a multi-band Landsat-5 TM satellite image composite.

Dataset / Tools: Ward/borough-level dengue and malaria case records for Kolkata and West Bengal, combined with satellite remote-sensing imagery.

Key Findings: The epidemic was found to affect the majority of the study area within roughly 11 days of onset, with disease clusters concentrated in specific central and southern boroughs; remote-sensing-derived environmental variables (vegetation, water bodies, land use) were shown to meaningfully improve outbreak-risk mapping.

Limitation: The Knox-test and kernel-density methods require dense, accurately geo-referenced case data, which is often unavailable in low-resource public-health reporting settings; the model is disease- and region-specific and requires re-calibration for other contexts.

Relevance to Assigned Problem: *Extends Objective 3 to fine spatial granularity, showing that ward/village-level GIS analysis combined with satellite-derived environmental covariates can localize hotspots precisely enough to support the resource-planning goals of Objective 6.*

4.8 [P8] Springer Study (2021) — Cloud and IoMT-Based Big Data Analytics System During the COVID-19 Pandemic

Objective: To propose a Cloud-Internet of Medical Things (IoMT) framework capable of efficiently collecting, transmitting, and analyzing large volumes of wearable-sensor health data generated during an infectious-disease outbreak.

Methodology: System-architecture design combining wearable sensors (temperature, glucose, heartbeat, chest/respiration sensors), IoMT gateways, and a cloud data-analytics layer; Hadoop MapReduce is proposed for parallel processing of the resulting big-data volumes.

Dataset / Tools: Illustrative wearable-sensor data streams generated during COVID-19 remote patient monitoring; no single benchmark dataset.

Key Findings: The proposed Cloud-IoMT architecture is shown, through system-level analysis, to be capable of handling the throughput and heterogeneity of multi-sensor pandemic monitoring data, supporting near-real-time analytics for clinical decision support.

Limitation: The paper does not report empirical latency, throughput, or accuracy benchmarks on a deployed system; resource requirements (bandwidth, storage cost) for large-scale rollout are only briefly discussed.

Relevance to Assigned Problem: *Informs Objective 1 (collect real-time public health data) and the underlying cloud infrastructure needed to support Objectives 6 and 7 (resource planning and analytics dashboards), by proposing a Cloud-IoMT big-data architecture capable of ingesting multi-sensor wearable data streams.*

4.9 [P9] PMC Study (2020) — An IoT-Based Framework for Early Identification and Monitoring of COVID-19 Cases

Source: PubMed Central (PMC).

Objective: To design an IoT-enabled framework that allows suspected COVID-19 patients to upload real-time symptom data, enabling machine-learning-based early identification and remote monitoring without requiring in-person clinical visits.

Methodology: A five-component architecture: symptom data collection/upload, a quarantine/isolation center, a data-analysis center, a physician interface, and a cloud infrastructure connecting all components; a machine-learning model classifies incoming symptom data to flag probable infection.

Dataset / Tools: Simulated/prototype symptom-reporting data used to validate the framework design; not a large-scale field deployment dataset.

Key Findings: The framework enables physicians to triage suspected cases remotely based on real-time uploaded symptom data, supporting faster isolation decisions and reducing unnecessary exposure at healthcare facilities.

Limitation: Relies on self-reported symptom data, which is susceptible to reporting bias and non-compliance; the framework was validated at prototype scale rather than in a large population deployment.

Relevance to Assigned Problem: *Supports Objective 1 and Objective 4 by demonstrating a working IoT-based symptom-collection and triage framework that enables real-time data capture and rapid early identification of probable cases, a direct precursor to automated alert generation.*

4.10 [P10] PMC Scoping Review (2025) — Early Warning of Infectious Disease Outbreaks Using Social Media and Digital Data: A Scoping Review

Objective: To systematically map and characterize the methodologies, performance, and limitations of digital-surveillance tools (social media, search-engine queries, participatory platforms) for early outbreak warning, relative to traditional epidemiological monitoring.

Methodology: A scoping review conducted per Joanna Briggs Institute and PRISMA-ScR guidelines across PubMed, Scopus, and Web of Science, comparing digital-source analytical techniques including ARIMA time-series models, supervised ML (SVM, LASSO, random forest), and NLP-based symptom/sentiment mining.

Dataset / Tools: Evidence synthesized from multiple published digital-surveillance studies using Google Trends, X (Twitter), and participatory-surveillance data.

Key Findings: Digital surveillance provided meaningful lead time (days to several weeks) ahead of official reporting in many reviewed studies, with correlations exceeding $r > 0.8$ against official case data for influenza and COVID-19 in several models; Google Trends and X were the most frequently used digital sources.

Limitation: Digital-surveillance data quality and demographic representativeness vary widely across platforms and regions, and the review calls for standardized reporting guidelines and bias-mitigation techniques (stratification, model weighting) before such tools can be relied upon as a primary surveillance channel.

Relevance to Assigned Problem: *Supports Objective 4 and Objective 5 (early warning alerts and monitoring transmission trends) by showing that non-clinical digital signals can provide multi-week lead time, making digital surveillance a valuable complementary data channel for the assigned platform's early-warning module.*

5. Comparative Analysis of Reviewed Literature

Table 1 consolidates the ten reviewed studies against a common set of parameters — research focus, methodology, dataset/tools used, key findings, and identified limitations — to enable a structured cross-paper comparison, as required for the assigned literature-survey deliverable.

Author/Year	Research Focus	Methodology	Dataset/Tools	Key Findings	Limitations / Research Gap
P1 Ekundayo, F. (2024)	Using Machine Learning to Predict Disease Outbreaks and Enhance Public...	A conceptual and literature-based review of supervised learning (regression, ensemble classifiers) and unsuper...	Synthesized from multiple secondary sources: EHR repositories, genomic databases, and social-me...	Supervised models were found effective for forecasting influenza-like illness trends, while unsupervised clustering was useful for...	Being a review rather than an empirical study, it does not report quantitative accuracy figures for a specific deployed ...
P2 SciRP Journal Study (2026)	Artificial Intelligence-Driven Approaches to Disease	Structured literature review classifying AI applications into four functional	Aggregated evidence drawn from published case studies of automated	AI-enabled automation of case detection and anomaly flagging	Provides limited discussion of computational cost, scalability testing, or

Author/Year	Research Focus	Methodology	Dataset/Tools	Key Findings	Limitations / Research Gap
	Surveillance and ...	categories: automated case detec...	surveillance systems; not ti...	reduces the lag between disease onset and official reporting. Predict...	interoperability with legacy hospital informa...
P3 PMC Study Group (2024–2025)	AI-Enabled Diagnostic Prediction within Electronic Health Records to E...	Retrospective cohort analysis using supervised ML models trained on historical EHR diagnostic codes; high-conf...	4.5 million de-identified patient records spanning 2010–2022, drawn from a large regional healt...	Among diseases with five or more recorded outbreaks between 2014 and 2022, the ML-augmented approach detected 33.3% of outbreaks e...	Performance is highly dependent on the availability of sufficient disease-specific historical training data; diseases wi...
P4 Ajeema Arsi, A. et al. (2025)	Harnessing Machine Learning and AI Techniques for Accurate Disease Out...	Comparative survey of forecasting models applied to medical records, meteorological data, and social signals, ...	Composite of medical records, weather/climate datasets, and social-media-derived signals report...	The combination of AI/ML/DL techniques with multi-modal data (clinical + environmental + social) consistently outperformed single-...	Data privacy, data quality inconsistency, and limited interpretability of deep-learning forecasts are identified as open...
P5 MDPI Study (2025)	AI-Assisted Real-Time Monitoring of Infectious Diseases in Urban Areas...	System design and simulation combining LSTM-based time-series forecasting, logistic regression for outbreak-pr...	Simulated/composite urban health data streams: wearable sensor feeds, pharmacy sales records, a...	The multi-source fusion architecture improved both the accuracy and timeliness of outbreak prediction relative to single-source mo...	Interoperability across heterogeneous device ecosystems, data-privacy compliance, and minimizing false-positive alerts r...

Author/Year	Research Focus	Methodology	Dataset/Tools	Key Findings	Limitations / Research Gap
P6 ResearchGate Study (2021)	GIS-Based Spatial Analysis and Hotspot Detection of COVID-19 Outbreak	Calculation of cumulative incidence rate (CIR) and cumulative fatality rate (CFR) per country; spatial-pattern...	Global, country-level COVID-19 case and fatality data collected from the outbreak onset through...	Significant spatial clustering of COVID-19 fatality was identified, with Yemen, Mexico, and Ecuador showing the highest case-fatal...	Country-level aggregation masks sub-national (city/village-level) variation and cannot support the fine-grained hotspot ...
P7 Kolkata/West Bengal GIS Study	Spatiotemporal Pattern and Hotspot Detection of Malaria and Dengue Usi...	Daily case-mapping over the outbreak period; spatio-temporal clustering using the Knox test; an environmental ...	Ward/borough-level dengue and malaria case records for Kolkata and West Bengal, combined with s...	The epidemic was found to affect the majority of the study area within roughly 11 days of onset, with disease clusters concentrate...	The Knox-test and kernel-density methods require dense, accurately geo-referenced case data, which is often unavailable ...
P8 Springer Study (2021)	Cloud and IoMT-Based Big Data Analytics System During the COVID-19 Pan...	System-architecture design combining wearable sensors (temperature, glucose, heartbeat, chest/respiration sens...	Illustrative wearable-sensor data streams generated during COVID-19 remote patient monitoring; ...	The proposed Cloud-IoMT architecture is shown, through system-level analysis, to be capable of handling the throughput and heterog...	The paper does not report empirical latency, throughput, or accuracy benchmarks on a deployed system; resource requireme...
P9 PMC Study (2020)	An IoT-Based Framework for Early Identification and Monitoring of COVI...	A five-component architecture: symptom data collection/upload, a quarantine/isolation center, a data-analysis ...	Simulated/prototype symptom-reporting data used to validate the framework design; not a large-s...	The framework enables physicians to triage suspected cases remotely based on real-time uploaded	Relies on self-reported symptom data, which is susceptible to reporting bias and non-compliance; the framework was valid...

Author/Year	Research Focus	Methodology	Dataset/Tools	Key Findings	Limitations / Research Gap
				symptom data, supporting faster is...	
P10 PMC Scoping Review (2025)	Early Warning of Infectious Disease Outbreaks Using Social Media and D...	A scoping review conducted per Joanna Briggs Institute and PRISMA-ScR guidelines across PubMed, Scopus, and We...	Evidence synthesized from multiple published digital-surveillance studies using Google Trends, ...	Digital surveillance provided meaningful lead time (days to several weeks) ahead of official reporting in many reviewed studies, w...	Digital-surveillance data quality and demographic representativeness vary widely across platforms and regions, and the r...

Table 1: Comparative summary of the reviewed literature on intelligent public health surveillance and outbreak prediction.

5.1 Comparison by Technology Dimension

The reviewed literature naturally clusters into the four technology dimensions named in the assigned problem statement:

- AI/ML-based prediction (P1–P5): These studies show consistently that fusing multiple heterogeneous data sources (EHR, climate, social, wearable) with supervised or deep-learning models improves both prediction accuracy and lead time over single-source statistical baselines, with P3 providing the strongest empirical evidence (up to 24-day earlier detection on real EHR data).
- GIS-based spatial epidemiology (P6–P7): These studies confirm that classical and modern spatial statistics (Moran's I, Getis-Ord Gi*, Knox test, kernel density estimation) reliably identify outbreak hotspots at both macro (country) and micro (ward/village) scale, but their resolution and usefulness depend heavily on the granularity and completeness of geo-referenced case data.
- IoT/Cloud-based real-time acquisition (P8–P9): These studies demonstrate architectural feasibility for continuous, sensor-driven health monitoring at scale, but both are limited to system-design or prototype-level validation rather than large-scale empirical deployment.
- Digital/social-media-based early warning (P10): This scoping review shows that non-clinical digital signals can provide multi-week lead time ahead of official reporting, but with notable data-quality and representativeness caveats.

Overall, the comparison indicates that no single reviewed study integrates all four dimensions into one unified, empirically validated, real-time platform — each study addresses one or two dimensions in isolation. This observation directly motivates the research gap discussed in Section 7.

5.2 Comparison by Performance-Related Parameters

Table 2 compares the reviewed studies against problem-specific performance parameters — detection lead time, spatial/analytical granularity, and scalability/validation maturity — since accuracy figures are not uniformly reported across all ten studies (several are literature reviews rather than empirical deployments).

Study	Detection Lead Time	Granularity	Validation Maturity
P1 – Ekundayo (2024)	Not quantified (review)	Population-level, multi-source	Conceptual review only
P2 – SCIRP (2026)	Not quantified (review)	Case/syndromic level	Conceptual review only
P3 – PMC EHR Study	1–24 days earlier (avg. lead time reported)	Individual patient / diagnosis level	Empirical, 4.5M records, 2010–2022
P4 – IJRASET (2025)	Not quantified (survey)	Multi-modal (clinical+climate+social)	Comparative survey only
P5 – SmartHealth-Track	Real-time (simulated)	Urban population level	Simulation / design study
P6 – GIS COVID-19	N/A (spatial pattern, not lead time)	Country level	Empirical, global dataset
P7 – Kolkata/WB GIS	N/A (spatial pattern, not lead time)	Ward/village level (fine-grained)	Empirical, regional dataset
P8 – Cloud-IoMT	Near real-time (architecture)	Individual sensor-stream level	Architecture proposal, no field benchmark
P9 – IoT COVID-19	Real-time symptom triage	Individual patient level	Prototype validation
P10 – Digital Scoping Review	Days to several weeks earlier	Population level (platform-dependent)	Systematic scoping review, $r > 0.8$ correlation reported

Table 2: Comparison of reviewed studies by detection lead time, analytical granularity, and validation maturity.

This performance-oriented comparison reinforces the observation in Section 5.1: empirically validated studies (P3, P6, P7, P9, P10) tend to focus on a single technology dimension at a fine level of maturity, whereas studies that attempt broader multi-dimension integration (P4, P5, P8) remain at the conceptual, comparative-survey, or architecture-proposal stage without full field validation. This trade-off between integration breadth and validation depth is precisely the gap that future work, as proposed in Section 8, needs to close.

6. Limitations and Challenges of Existing Approaches

Synthesizing the individual and comparative review above, the following recurring limitations and challenges emerge across the existing body of literature relevant to the assigned problem statement:

6.1 Data-Related Challenges

- Data fragmentation: Health data remains siloed across hospitals, laboratories, and public-health agencies, with few reviewed systems demonstrating true multi-source integration (P1, P2, P4).
- Data quality and completeness: ML-based prediction accuracy depends heavily on sufficient historical, disease-specific training data; diseases with sparse records show weaker predictive performance (P3).
- Geo-referencing granularity: Spatial hotspot detection is only as fine-grained as the underlying case geo-location data; coarse country-level data masks locally actionable clusters (P6 vs. P7).
- Representativeness bias in digital sources: Social-media- and search-based signals over-represent digitally active, urban populations and under-represent regions with limited internet access (P10).

6.2 Technical and Architectural Challenges

- Limited empirical validation at scale: Several IoT/cloud architectures (P8, P9) are validated only at prototype or simulation level rather than in large-scale field deployment.
- Interoperability: Heterogeneous device ecosystems, EHR standards, and reporting formats hinder seamless data exchange between hospitals, IoT devices, and analytics platforms (P4, P5).
- False-positive / alert-fatigue risk: Real-time anomaly-detection systems must balance sensitivity against the operational cost of excessive false alerts (P2, P5).
- Model interpretability: Deep-learning-based forecasting models are frequently criticized for being "black boxes," limiting public-health officials' trust and actionable use of their outputs (P1, P4).

6.3 Governance and Ethical Challenges

- Privacy and data-protection compliance: Continuous collection of clinical, location, and wearable-sensor data raises significant privacy concerns that are acknowledged but not fully resolved in the reviewed literature.
- Lack of standardized evaluation: No consistent benchmark or reporting standard exists across studies, making it difficult to compare accuracy, lead time, or false-positive rates on an equal basis (P10).

7. Research Gap

Based on the comparative analysis in Section 5 and the limitations identified in Section 6, the following research gap is identified with respect to the assigned problem statement of building an intelligent public health surveillance and disease outbreak prediction system:

Existing research addresses the four required technology components — AI-based prediction, GIS-based spatial hotspot detection, IoT/cloud-based real-time data acquisition, and digital/social-signal-based early warning — largely in isolation, with each reviewed study contributing to only one or two of these dimensions. There is no reviewed study that presents a single, empirically validated, end-to-end platform that (a) ingests real-time multi-source data from hospitals, laboratories, IoT/wearable devices, and digital/social channels; (b) applies AI-based predictive analytics with interpretable outputs; (c) performs fine-grained GIS-based hotspot identification; and (d) delivers this through a unified epidemiological dashboard that supports both automated early-warning alerts and resource-planning decision support for public-health administrators.

In addition, a secondary research gap concerns evaluation standardization: because the reviewed studies use different datasets, geographic scopes, and performance metrics, there is no common benchmark against which an integrated system's accuracy, lead time, and false-positive rate can be objectively measured and compared against the existing single-dimension approaches.

8. Future Research Directions and Proposed Improvements

Guided by the identified research gap and the objectives of the assigned problem statement, the following future research directions and system-level improvements are proposed:

8.1 Unified Multi-Source Data Integration Layer

Future work should design a common data-ingestion and normalization layer capable of consuming structured EHR/laboratory feeds, IoT/wearable sensor streams, GIS-tagged case records, and unstructured digital signals (social media, search trends) into a single interoperable data model, addressing the fragmentation limitation noted in Section 6.1.

8.2 Explainable, Multi-Model AI Prediction Engine

Rather than relying on a single algorithm, an ensemble of interpretable models (e.g., gradient-boosted trees combined with attention-based deep-learning components, augmented with explainable-AI techniques such as SHAP) should be used to generate outbreak-risk scores that public-health staff can trust and act upon, directly addressing objective 2 (predict disease outbreaks using AI) of the assigned problem statement.

8.3 Fine-Grained GIS Hotspot Engine

Combining ward/village-level geo-tagged case data with satellite-derived environmental covariates (as demonstrated in P7) and modern spatial-statistics techniques (Getis-Ord G_i^* , kernel density estimation) can

extend the macro-level hotspot detection shown in P6 down to actionable, localized resolution, addressing objective 3 (identify outbreak hotspots through GIS).

8.4 Edge-Enabled Real-Time IoT/Cloud Pipeline

Deploying lightweight, edge-compatible inference models close to IoT/wearable data sources (as suggested in P5) — combined with elastic cloud back-ends for heavier batch analytics (as proposed in P8) — can reduce end-to-end latency while remaining scalable, addressing objective 1 (collect real-time public health data) and supporting objective 4 (generate early warning alerts).

8.5 Standardized Evaluation Benchmark

The research community would benefit from a shared, de-identified benchmark dataset and a standardized set of evaluation metrics (lead time, sensitivity, false-positive rate per year, spatial resolution) so that future integrated systems can be objectively compared, addressing the secondary research gap identified in Section 7.

8.6 Decision-Support Dashboard for Public Health Administrators

Finally, translating model outputs into an accessible epidemiological dashboard — with hotspot maps, transmission-trend charts, and resource-planning recommendations — is essential to fulfil objectives 6 through 8 of the assigned problem statement (support healthcare resource planning, provide epidemiological analytics dashboards, and improve public health decision-making), and would directly enable the expected outcomes of early detection, improved surveillance, and increased community health protection listed in the problem statement.

8.7 Outline of a Proposed Integrated System Architecture

Building on the future directions above, this subsection outlines a conceptual four-layer architecture that synthesizes the strengths of the reviewed literature into a single system directly aligned with the assigned problem statement's objectives and expected outcomes. This outline is intended as a design reference for future implementation work rather than as a report of an already-built system.

Layer 1 — Data Acquisition Layer

This layer is responsible for Objective 1 (collect real-time public health data). It aggregates structured data from hospital and laboratory information systems, unstructured clinical notes, IoT/wearable sensor feeds (as in P5, P8, P9), GIS-tagged case locations, and digital signals such as search-trend and social-media data (as in P10). A common health-data interoperability standard (e.g., HL7 FHIR) is recommended for structured feeds, while a lightweight message-queue (e.g., Kafka-style streaming, as noted in the real-time surveillance literature) handles high-velocity sensor and digital-signal ingestion.

Layer 2 — Analytics and Prediction Layer

This layer addresses Objectives 2 and 5 (predict disease outbreaks using AI; monitor disease transmission trends). It hosts an ensemble of interpretable machine-learning and deep-learning models — time-series forecasters (LSTM, as in P5), anomaly-detection models (isolation forests, as in P5), and diagnostic-prediction classifiers (as

in P3) — supplemented with explainable-AI overlays (attention weights, Shapley values, as recommended in P1) so that outputs remain auditable by public-health staff. In parallel, a GIS analytics module (Objective 3) performs spatial hotspot detection using Getis-Ord G_i^* , kernel density estimation, and the Knox test (as in P6 and P7), enriched with remote-sensing environmental covariates where available.

Layer 3 — Alerting and Decision-Support Layer

This layer addresses Objective 4 (generate early warning alerts) and Objective 6 (support healthcare resource planning). Outbreak-risk scores and hotspot maps generated in Layer 2 are passed through a configurable thresholding and alert-fatigue-management module, informed by the false-positive-rate benchmarks reported in P3 (approximately 1.33 false alerts per year), before being routed to public-health administrators and, where appropriate, to affected facilities for resource pre-positioning.

Layer 4 — Visualization and Governance Layer

This layer addresses Objectives 7 and 8 (provide epidemiological analytics dashboards; improve public health decision-making). It presents hotspot maps, transmission-trend charts, and resource-allocation recommendations through a role-based dashboard, and incorporates the data-governance, privacy, and standardized-evaluation safeguards discussed in Sections 6 and 8.5 to ensure the system is deployable in a manner consistent with data-protection regulations and ethical-AI principles.

Together, these four layers map directly onto the eight objectives and eight expected outcomes stated in the assigned problem statement, and represent a concrete architectural response to the research gap identified in Section 7.

9. Conclusion

This literature survey reviewed ten research contributions spanning AI/machine-learning-based outbreak prediction, GIS-based spatial epidemiology, IoT/cloud-based real-time health monitoring, and digital/social-media-based early warning, in direct response to the assigned problem statement on intelligent public health surveillance. The comparative analysis showed that while each technology dimension has been independently validated with promising results — including up to 24-day earlier outbreak detection using EHR-based ML models, statistically significant spatial hotspot identification using GIS techniques, and multi-week lead time using digital-surveillance signals — no single reviewed system integrates all these dimensions into one empirically validated, real-time, decision-support platform. The identified research gap therefore centers on the absence of a unified, interoperable, and explainable system that combines real-time multi-source data ingestion, AI-based prediction, fine-grained GIS hotspot detection, and administrator-facing decision-support dashboards. Addressing this gap, as outlined in the proposed future directions, would substantially advance the objectives and expected outcomes defined in the assigned problem statement, ultimately contributing to faster outbreak detection, more efficient healthcare resource allocation, and stronger community health protection.

10. References

The following references are formatted in IEEE citation style, consistent with the referencing guideline specified in the assessment instructions.

- [1] F. Ekundayo, "Using machine learning to predict disease outbreaks and enhance public health surveillance," *World Journal of Advanced Research and Reviews*, vol. 24, no. 3, pp. 794–811, 2024.
- [2] "Artificial Intelligence-Driven Approaches to Disease Surveillance and Outbreak Prediction," *Scientific Research Publishing (SCIRP) Open Access Journal*, 2026.
- [3] "AI-Enabled Diagnostic Prediction within Electronic Health Records to Enhance Biosurveillance and Early Outbreak Detection," *PubMed Central (PMC)*, Article ID PMC12204487.
- [4] A. Ajeema Arsi, J. Daphine Joel, B. Breenu Bhasker, A. Agnel Macdalin, and H. S. Abinaya Hari, "Harnessing machine learning and AI techniques for accurate disease outbreak forecasting," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, 2025, doi: 10.22214/ijraset.2025.73613.
- [5] "AI-Assisted Real-Time Monitoring of Infectious Diseases in Urban Areas (SmartHealth-Track)," *MDPI Mathematics*, vol. 13, no. 12, p. 1911, 2025.
- [6] "GIS-Based Spatial Analysis and Hotspot Detection of COVID-19 Outbreak," *ResearchGate*, 2021.
- [7] "Spatiotemporal Pattern and Hotspot Detection of Malaria Using Spatial Analysis and GIS in West Bengal: An Approach to Medical GIS," *ResearchGate*, 2020.
- [8] "Cloud and IoMT-Based Big Data Analytics System During COVID-19 Pandemic," in *Big Data Analytics and Computational Intelligence for Cybersecurity*, Springer, 2021.
- [9] "An IoT-Based Framework for Early Identification and Monitoring of COVID-19 Cases," *PubMed Central (PMC)*, Article ID PMC7428786, 2020.
- [10] "Early Warning of Infectious Disease Outbreaks Using Social Media and Digital Data: A Scoping Review," *PubMed Central (PMC) / PubMed*, 2025.
- [11] "Spatio-Temporal Diffusion Pattern and Hotspot Detection of Dengue in Chachoengsao Province, Thailand," *PubMed Central (PMC)*, Article ID PMC3037060.
- [12] "Digital Epidemiology: Harnessing Big Data for Early Detection and Monitoring of Viral Outbreaks," *ScienceDirect*, 2024.

Annexure A — Glossary of Key Terms

Term	Definition
AI (Artificial Intelligence)	The broad field of computer science concerned with building systems capable of performing tasks that normally require human intelligence, such as pattern recognition and prediction.
Machine Learning (ML)	A subset of AI in which models learn patterns directly from data rather than being explicitly programmed with rules.
Deep Learning (DL)	A subset of ML using multi-layered neural networks, well suited to complex, high-dimensional data such as time series or text.
LSTM (Long Short-Term Memory)	A type of recurrent neural network architecture designed to model sequential/time-series data, commonly used for outbreak-trend forecasting.
NLP (Natural Language Processing)	AI techniques for extracting meaning from unstructured text, such as clinical notes or social-media posts.
GIS (Geographic Information System)	A system for capturing, storing, analyzing, and visualizing spatially referenced data, used here to map and detect disease hotspots.
Getis-Ord G_i^*	A spatial statistic used to identify statistically significant clusters of high or low values (hotspots/coldspots) within a geographic dataset.
Moran's I	A measure of spatial autocorrelation indicating whether nearby geographic locations have similar values.
Kernel Density Estimation (KDE)	A statistical technique that converts discrete case-location points into a smooth, continuous risk-density surface.
IoT (Internet of Things)	A network of physical devices (sensors, wearables) that collect and transmit data over the internet.
IoMT (Internet of Medical Things)	The subset of IoT devices specifically used in healthcare and clinical monitoring contexts.
Edge/Fog Computing	Computing paradigms that perform data processing closer to the data source (rather than solely in a centralized cloud) to reduce latency.
EHR (Electronic Health Record)	A digital record of a patient's medical history maintained by a healthcare provider.
Digital Epidemiology	The use of non-clinical digital data sources, such as search-engine queries and social media, to monitor and predict disease activity.
Explainable AI (XAI)	Techniques (e.g., SHAP, attention mechanisms) that make the outputs of complex AI models interpretable to human users.